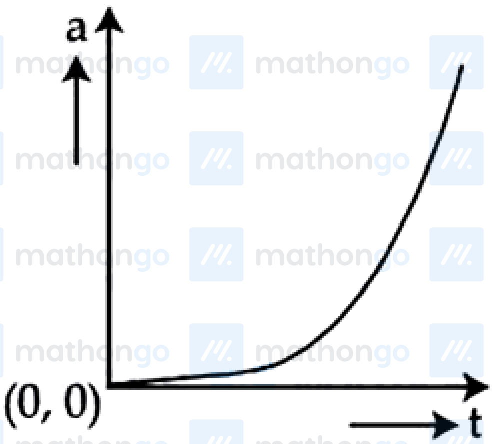
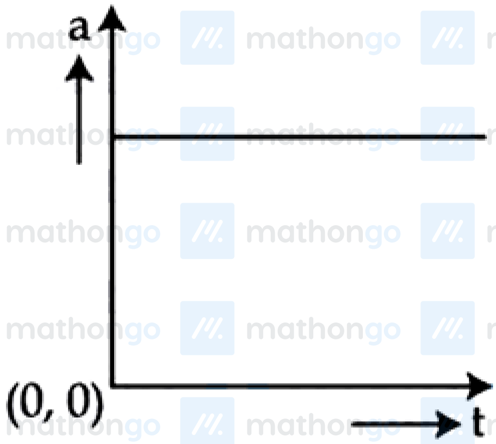
Q1 - 2024 (04 Apr Shift 1)

A wooden block, initially at rest on the ground, is pushed by a force which increases linearly with time t.

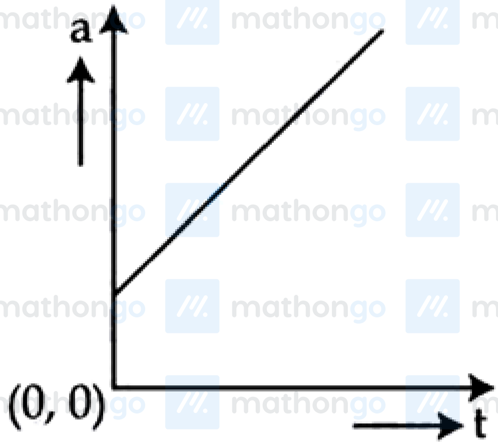
Which of the following curve best describes acceleration of the block with time:



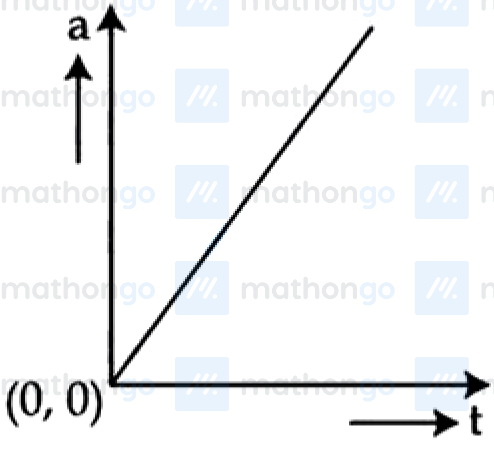
(1)



(2)



(3)

(4) 

Q2 - 2024 (04 Apr Shift 2)

A 2 kg brick begins to slide over a surface which is inclined at an angle of 45° with respect to horizontal axis. The co-efficient of static friction between their surfaces is:

(1) 1.7

(2) $\frac{1}{\sqrt{3}}$

(3) 0.5

(4) 1

Q3 - 2024 (05 Apr Shift 1)

A wooden block of mass 5 kg rests on a soft horizontal floor. When an iron cylinder of mass 25 kg is placed on the top of the block, the floor yields and the block and the cylinder together go down with an acceleration of $0.1 \mathrm{~ms}^{-2}$. The action force of the system on the floor is equal to:

(1) 196 N

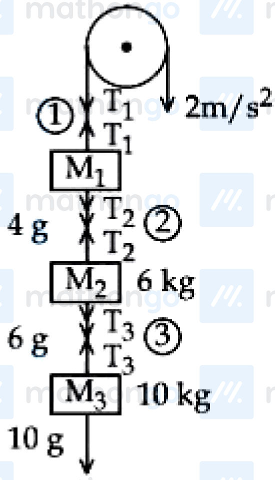
(2) 291 N

(3) 294 N

(4) 297 N

Q4 - 2024 (05 Apr Shift 1)

Three blocks M_1, M_2, M_3 having masses 4 kg, 6 kg and 10 kg respectively are hanging from a smooth pully using rope 1,2 and 3 as shown in figure. The tension in the rope $1, T_1$ when they are moving upward with acceleration of $2 \mathrm{~ms}^{-2}$ is _____ N (if $\mathrm{g}=10 \mathrm{~m/s}^2$).



Q5 - 2024 (05 Apr Shift 2)

A heavy box of mass 50 kg is moving on a horizontal surface. If co-efficient of kinetic friction between the box and horizontal surface is 0.3 then force of kinetic friction is :

(1) 1.47 N

(2) 147 N

(3) 14.7 N

(4) 1470 N

Q6 - 2024 (06 Apr Shift 1)

A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (where $m_2 > m_1$). If the acceleration of the system is $\frac{g}{\sqrt{2}}$, then the ratio of the masses $\frac{m_1}{m_2}$ is:

(1) $\frac{1+\sqrt{5}}{\sqrt{5}-1}$

(2) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

(3) $\frac{1+\sqrt{5}}{\sqrt{2}-1}$

(4) $\frac{\sqrt{3}+1}{\sqrt{2}-1}$

Q7 - 2024 (06 Apr Shift 2)

A body of weight 200 N is suspended from a tree branch through a chain of mass 10 kg. The branch pulls the chain by a force equal to (if $g = 10\ \mathrm{m/s^2}$) :

(1) 100 N

(2) 200 N

(3) 300 N

(4) 150 N

Q8 - 2024 (06 Apr Shift 2)

A car of 800 kg is taking turn on a banked road of radius 300 m and angle of banking 30°. If coefficient of static friction is 0.2 then the maximum speed with which car can negotiate the turn safely:
$\left(\mathrm{g} = 10\ \mathrm{m/s^2}, \sqrt{3} = 1.73\right)$

(1) $264\ \mathrm{m/s}$

(2) $51.4\ \mathrm{m/s}$

(3) $70.4\ \mathrm{m/s}$

(4) $102.8\ \mathrm{m/s}$

Q9 - 2024 (08 Apr Shift 2)

A given object takes n times the time to slide down 45° rough inclined plane as it takes the time to slide down an identical perfectly smooth 45° inclined plane. The coefficient of kinetic friction between the object and the surface of inclined plane is :

(1) $\sqrt{1-\frac{1}{n^2}}$

(2) $1-n^2$

(3) $1-\frac{1}{n^2}$

(4) $\sqrt{1-n^2}$

Q10 - 2024 (09 Apr Shift 1)

A light unstretchable string passing over a smooth light pulley connects two blocks of masses m_1 and m_2. If the acceleration of the system is $\frac{g}{8}$, then the ratio of the masses $\frac{m_2}{m_1}$ is :

(1) 8 : 1

(2) 5 : 3

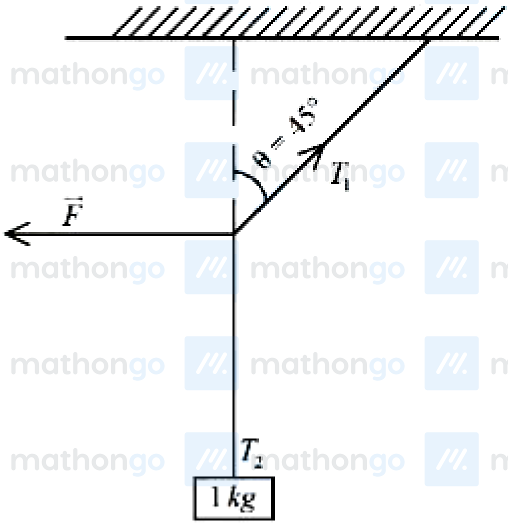
(3) 4 : 3

(4) 9 : 7

Q11 - 2024 (09 Apr Shift 2)

A 1 kg mass is suspended from the ceiling by a rope of length 4 m. A horizontal force ' F ' is applied at the mid point of the rope so that the rope makes an angle of 45° with respect to the vertical axis as shown in figure. The magnitude of F is :

(Assume that the system is in equilibrium and $g=10 \mathrm{~m} / \mathrm{s}^2$)



(1) 10 N

(2) $\frac{10}{\sqrt{2}}$ N

(3) 1 N

(4) $\frac{1}{10 \times \sqrt{2}} N$

Answer Key

Q1 (4) **Q2** (4) **Q3** (2) **Q4** (240)

Q5 (2) **Q6** (2) **Q7** (3) **Q8** (2)

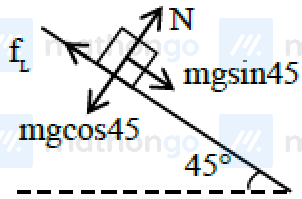
Q9 (3) **Q10** (4) **Q11** (1)

Q1

$\mathrm{F}=\mathrm{ma} \Rightarrow \mathrm{a}=\frac{\mathrm{F}}{\mathrm{m}}=\frac{\mathrm{kt}}{\mathrm{m}}$

a vs t will be straight line passing through origin.

Q2



$\mathrm{mg} \sin 45=\mathrm{f}_{\mathrm{L}}$
$\mathrm{mg} \cos 45=\mathrm{N}$
$\mathrm{f}_{\mathrm{L}}=\mu_{\mathrm{s}} N$
$\mu_{\mathrm{s}}=\tan 45=1$

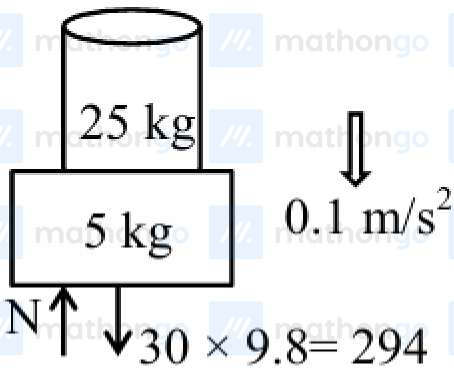
or

$\tan \theta=\mu_s$ (θ is angle of repose)

$\tan 45=\mu_{\mathrm{s}}=1$

Q3

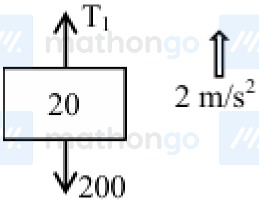
Taking $\mathrm{g}=9.8 \mathrm{~m} / \mathrm{s}^2$



$294-\mathrm{N}=30 \times 0.1$
$\mathrm{N}=291$

Q4

FBD of M_1 :



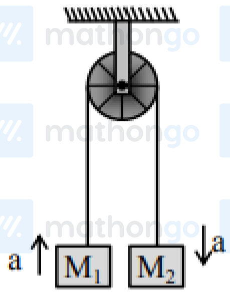
$T_1 - 200 = (4 + 6 + 10) \times 2$

$\therefore T_1 = 240$

Q5

$F_k = \mu_k N = 0.3 \times 50 \times 9.8 = 147N$

Q6



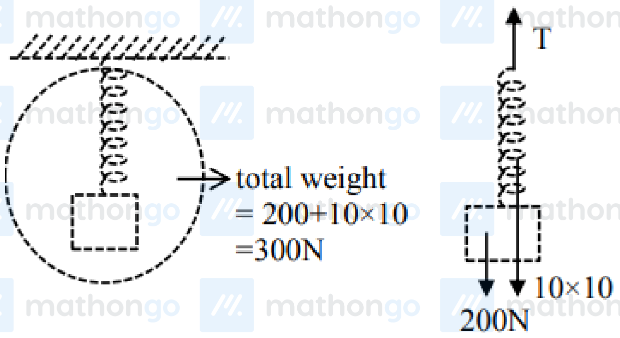
$$a = \left(\frac{M_2 - M_1}{M_1 + M_2}\right) g$$

$$\frac{g}{\sqrt{2}} = \left(\frac{M_2 - M_1}{M_1 + M_2}\right) g$$

$$(M_1 + M_2) = \sqrt{2}M_2 - \sqrt{2}M_1$$

$$\frac{M_1}{M_2} = \left(\frac{\sqrt{2} - 1}{\sqrt{2} + 1}\right)$$

Q7



Chain block system is in equilibrium so

$T = 200 + 100 = 300$ N.

Q8

$m = 800$ kg
$r = 300$ m
$\theta = 30^\circ$
$\mu_s = 0.2$

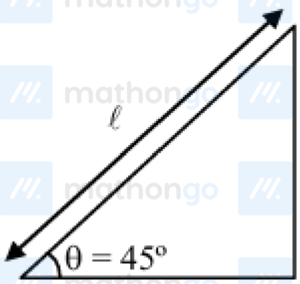
$$V_{max} = \sqrt{Rg\left[\frac{\tan\theta + \mu}{1 - \mu\tan\theta}\right]}$$

$$= \sqrt{300 \times g \times \left[\frac{\tan 30^\circ + 0.2}{1 - 0.2 \times \tan 30^\circ}\right]}$$

$$= \sqrt{300 \times 10 \times \left[\frac{0.57 + 0.2}{1 - 0.2 \times 0.57}\right]}$$

$V_{max} = 51.4$ m/s

Q9



Case-1 : No friction

$a = g\sin\theta$

$\ell = \frac{1}{2}(g\sin\theta)t_1^2$

$t_1 = \sqrt{\frac{2\ell}{g\sin\theta}}$

Case-2 : With friction

$a = g\sin\theta - \mu g\cos\theta$

$\ell = \frac{1}{2}(g\sin\theta - \mu g\cos\theta)t_2^2$

$\sqrt{\frac{2\ell}{g\sin\theta - \mu g\cos\theta}} = n\sqrt{\frac{2\ell}{g\sin\theta}}$

$\mu = 1 - \frac{1}{n^2}$

Q10

$a_{sys} = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)g = \frac{g}{8}$

$\Rightarrow \frac{m_2}{m_1} = \frac{9}{7}$

Q11

$T_1\sin 45^\circ = F$

$T_1\cos 45^\circ = T_2 = 1 \times g$

$\therefore \tan 45^\circ = \frac{F}{g}$

$\therefore F = 10\ N$